Assignment set II

Sunday, May 31, 2026 8:53 AM

1.1) $\dot{x} = rx(1-\frac{x}{K}) - H$ $\quad r, K, H > 0,\ x \geq 0$

a) $H < \frac{rK}{4}$

lets find FP

$\dot{x} = rx(1-\frac{x}{K}) - H = 0$

$\frac{x^2}{K} - x + \frac{H}{r} = 0$

$x_{1,2} = \frac{1 \pm \sqrt{1-4\frac{H}{rK}}}{\frac{2}{K}} = \frac{K}{2}\left(1 \pm \sqrt{1-\frac{4H}{rK}}\right)$

this is positive because $H < \frac{rK}{4}$, $4H < rK$, $\frac{4H}{rK} < 1$

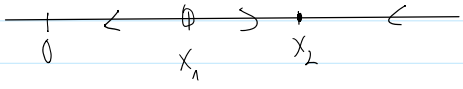
so $x_1 = \frac{K}{2}\left(1-\sqrt{1-\frac{4H}{rK}}\right)$

$x_2 = \frac{K}{2}\left(1+\sqrt{1-\frac{4H}{rK}}\right)$ and $x_1 < x_2$

for $0 < x < x_1$

lets check at $x = 0$

$\dot{x} = 0 - H < 0$ by continuity its true for the whole range



for $x_1 < x < x_2$

we can look at the root form

$\dot{x} = -\frac{rx^2}{K} + rx - H = -\frac{r}{K}(x-x_1)(x-x_2) = (-)(+)(-) = (+) > 0$

for $x > x_2$

$\dot{x} = -\frac{r}{K}(x-x_1)(x-x_2) = (-)(+)(+) = (-) < 0$

so x_1 is unstable FP, x_2 is stable FP

for any $x_1 \leq x_0$, the population never dies out

b) if $H > \frac{rK}{4}$ then $\frac{4H}{rK} < 1$

so $\dot{x} = 0$ has no real solutions

at $x = 0$, $\dot{x} = -H$ which is still negative

by continuity this true for every x

so if $H > \frac{rK}{4}$, $x(t)$ always decays to 0

1.2) $\dot{x} = x^2 - y$
$\dot{y} = x - y$

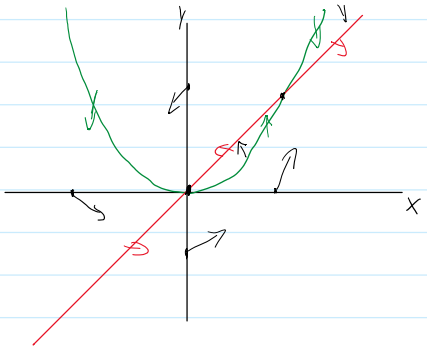
(a) FP where $\dot{y} = \dot{x} = 0$

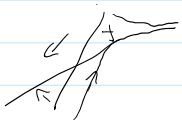
$x^2 - y = x - y$
$x(x-1) = 0$ $\quad x = 0, \; x = 1$
$\quad\;\; y = 0, \; y = 1$
FP are $(0,0), (1,1)$

(b) nullcline where $\dot{x} = 0$ or $\dot{y} = 0$

$\dot{x} = 0, \; y = x^2$
$\dot{y} = 0, \; y = x$
check directions on $y = x^2$

$\dot{y} = x - x^2$: $x < 0, \; \dot{y} < 0$
$0 < x < 1, \; \dot{y} > 0$
$x > 1, \; \dot{y} < 0$





now for $y = x$

$\dot{x} = y^2 - y$: $y < 0, \; \dot{x} > 0$
$0 < y < 1, \; \dot{x} < 0$
$y > 1, \; \dot{x} > 0$

(c) let's classify FPs

$DF\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x & -1 \\ 1 & -1 \end{pmatrix}$

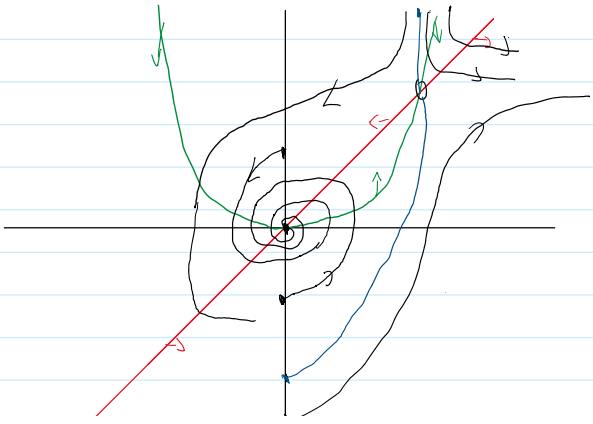
FP $(0,0)$ - $DF\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$

$\det DF\begin{pmatrix} 0 \\ 0 \end{pmatrix} = 1 > 0 \Rightarrow$ stable
$\operatorname{tr} DF\begin{pmatrix} 0 \\ 0 \end{pmatrix} = -1 < 0$
$(-1)^2 - 4 \cdot 1 < 0 \Rightarrow$ spiral

so $(0,0)$ is a spiral sink

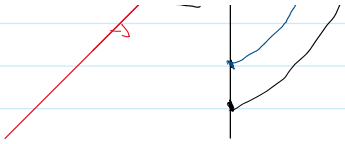
$DF\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & -1 \end{pmatrix}$

$\det DF\begin{pmatrix} 1 \\ 1 \end{pmatrix} = -1 < 0 \Rightarrow$ saddle

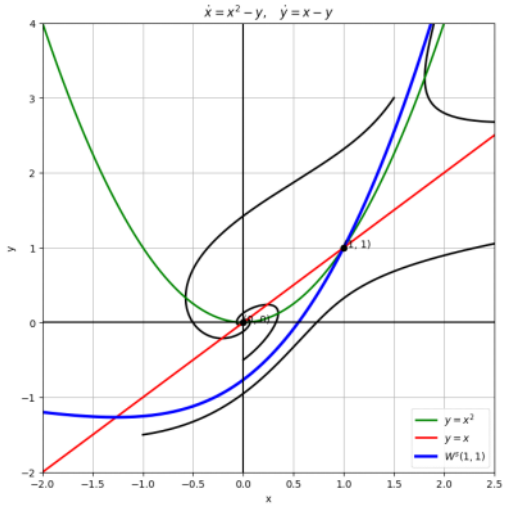


lines in blue are stable manifolds

stable manifolds



d)



1.3)

$$\dot{X} = X(K + aY - cX)$$
$$\dot{Y} = Y(L + bX - eY)$$

a) lets look for FPs $\dot{X} = \dot{Y} = 0$

$$X(K + aY - cX) = 0$$
$$Y(L + bX - eY) = 0$$

since we are looking for $X^*, Y^* > 0$, $X, Y \neq 0$

$$-cX + aY = -K$$
$$bX - eY = -L$$

$$\underbrace{\begin{pmatrix} -c & a \\ b & -e \end{pmatrix}}_{A} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} -K \\ -L \end{pmatrix}$$

$$\det A = ec - ab$$

assuming $ec \neq ab$, we can use cramer's rule

$$X^* = \frac{\det \begin{pmatrix} -K & a \\ -L & -e \end{pmatrix}}{\det A} = \frac{Ke + aL}{ec - ab}$$

$$Y^* = \frac{\det \begin{pmatrix} -c & -K \\ b & -L \end{pmatrix}}{\det A} = \frac{cL + Kb}{ec - ab}$$

check $X^*, Y^* > 0$, assuming $a, b, c, e, K, L \geq 0$

$$X^*, Y^* > 0 \iff ec > ab$$

(b) $DF\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} K+ay-2cx & ax \\ by & L+bx-2ey \end{pmatrix}$

at (x^*,y^*), $\dot{x}=\dot{y}=0$

so $0 = x(K+ay^*-cx^*)$ $\quad 0 = y(L+bx^*-ey^*)$

$0 = K+ay^*-cx^*$ $\quad 0 = L+bx^*-ey^*$

so

$$DF\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} \overbrace{K+ay^*-cx^*}^{0}-cx^* & ax^* \\ by^* & \underbrace{L+bx^*-ey^*}_{0}-ey^* \end{pmatrix} = \begin{pmatrix} -cx^* & ax^* \\ by^* & -ey^* \end{pmatrix}$$

$\det DF\begin{pmatrix} x^* \\ y^* \end{pmatrix} = cx^*ey^* - ax^*by^* = x^*y^*(ce-ab)$

$\operatorname{tr} DF\begin{pmatrix} x^* \\ y^* \end{pmatrix} = -cx^*-ey^*$

so if $ce > ab \Rightarrow x^*, y^* > 0 \Rightarrow \det DF\begin{pmatrix} x^* \\ y^* \end{pmatrix} > 0$ and $\operatorname{tr} DF\begin{pmatrix} x^* \\ y^* \end{pmatrix} < 0$

so (x^*, y^*) is stable $\qquad \frac{c}{a} > \frac{b}{e}$

(c) the nullclines are

$\dot{x}=0$: $x=0$, $\quad y = \frac{c}{a}x - \frac{K}{a}$

$\dot{y}=0$: $y=0$, $\quad y = \frac{b}{e}x + \frac{L}{e}$

check for direction on nullcline

for $x=0$, $\dot{y} = y(L-ey)$

$y > \frac{L}{e} \Rightarrow ey > L \Rightarrow \dot{y} < 0$

$0 < y < \frac{L}{e} \Rightarrow ey < L \Rightarrow \dot{y} > 0$

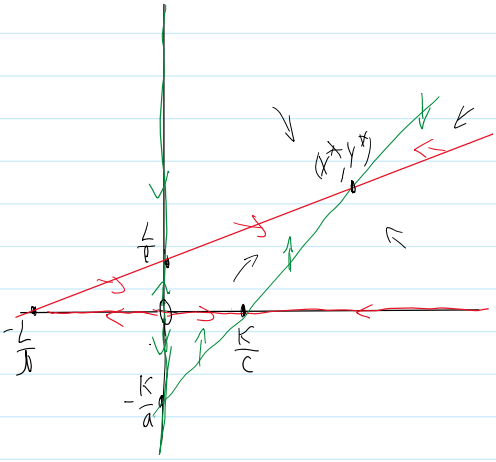
$y < 0 \Rightarrow \dot{y} < 0$

for $y=0$, $\dot{x} = x(K-cx)$

$x > \frac{K}{c} \Rightarrow cx > K \Rightarrow \dot{x} < 0$

$0 < x < \frac{K}{c} \Rightarrow cx < K \Rightarrow \dot{x} > 0$

$x < 0 \Rightarrow \dot{x} < 0$



for $x > 0$ $\quad \dot{x} > 0 \iff K+ay-cx > 0$

$y > \frac{c}{a}x - \frac{K}{a}$

so above this nullcline, $\dot{x} > 0$

below it $\dot{x} < 0$

for $y > 0$, $\quad \dot{y} > 0 \iff L+bx-ey > 0$

$y < \frac{b}{e}x + \frac{L}{e}$

so below this nullcline, $y > 0$

above it $\dot{y} > 0$

lets check other FP types

let's check other FP types

FPs are $(\frac{K}{c}, 0)$, $(0, \frac{L}{e})$, $(0,0)$, (x^*, y^*)

(x^*, y^*) is stable

$$DF\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} K + ay - 2cx & ax \\ by & L + bx - 2ey \end{pmatrix}$$

$$DF\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} K & 0 \\ 0 & L \end{pmatrix} \quad \begin{matrix} \det > 0 \\ tr > 0 \end{matrix} \quad \text{unstable}$$

$$DF\begin{pmatrix} \frac{K}{c} \\ 0 \end{pmatrix} = \begin{pmatrix} -K & \frac{aK}{c} \\ 0 & L + \frac{bK}{c} \end{pmatrix} \quad \det = -KL - \frac{bK^2}{c} < 0 \quad \text{saddle}$$

$$DF\begin{pmatrix} 0 \\ \frac{L}{e} \end{pmatrix} = \begin{pmatrix} K + \frac{aL}{e} & 0 \\ \frac{bL}{e} & -L \end{pmatrix} \quad \det = -LK - \frac{aL^2}{e} < 0 \quad \text{saddle}$$

the basin of attraction of (x^*, y^*) is any $\{(x,y) \mid x>0, y>0\}$.
any point on $x=0$ goes directly to $(0, \frac{L}{e})$ and
any point on $y=0$ goes directly to $(\frac{K}{c}, 0)$, so trajectories can't cross the axis.
the stable manifolds of the saddle points is the axis itself, so any other starting point is repelled.

for FP $(0,0)$ since it's unstable $W^s(0,0) = \{(0,0)\}$ and $W^u = \{(x,y) \mid x \neq 0 \text{ and } y \neq 0\}$

1.4

$$\dot{x} = -\beta xy$$
$$\dot{y} = \beta xy - \gamma y \qquad \text{assuming } \beta, \gamma > 0$$

(a) for $\dot{x} = 0 = -\beta xy$ $\quad x = 0$ or $y = 0$
for $\dot{y} = 0 = \beta xy - \gamma y = y(\beta x - \gamma)$, $\quad y = 0$ or $x = \frac{\gamma}{\beta}$

intersecting these FP are $\{(x,0) \mid x \in \mathbb{R}\}$
and the null clines are $x = 0, y = 0$ for $\dot{x}$, and $y = 0, x = \frac{\gamma}{\beta}$ for $\dot{y}$

(b)

$$\frac{\dot{y}}{\dot{x}} = \frac{\beta xy - \gamma y}{-\beta xy} = -1 + \frac{\gamma}{\beta x}$$

$$\dot{y} = \left(-1 + \frac{\gamma}{\beta x}\right)\dot{x} \qquad \text{integrating both sides}$$

$$y = -x + \frac{\gamma}{\beta} \ln x + C$$

$$y + x - \frac{\gamma}{\beta} \ln x = C$$

so $G(x,y) = y + x - \frac{\gamma}{\beta} \ln x$ is constant along trajectories.

$y + x - \frac{\gamma}{\beta} \ln x = C$

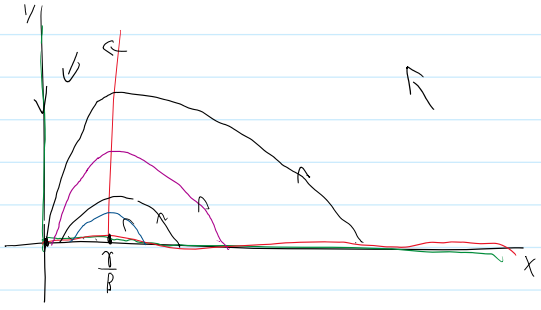
So $G(x,y) = y + x - \frac{\gamma}{\beta} \ln x$ is constant along trajectories.

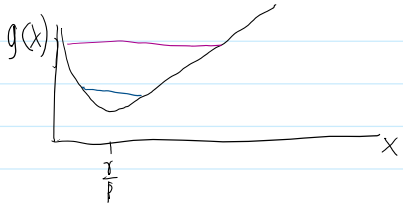
to make sure, we can check $\dot{G}$

$\dot{G} = \dot{y} + \dot{x} - \frac{\gamma}{\beta x} \cdot \dot{x} = \beta xy - \gamma y - \beta xy - \frac{\gamma \cdot \beta xy}{\beta x} = -\gamma y + \gamma y = 0$

(c) $G = y + x - \frac{\gamma}{\beta} \ln x = C$

$y = C - \underbrace{x + \frac{\gamma}{\beta} \ln x}_{g(x)}$





1.5 $\dot{x} = y$

$\dot{y} = -x + x^3$

(a) $\dot{x} = y = 0$

$\dot{y} = -x + x^3 = x(-1 + x^2) = 0$, $x = 0$ $\quad -1 + x^2 = 0$

$x = \pm 1$

so FP are $(0,0), (1,0), (-1,0)$

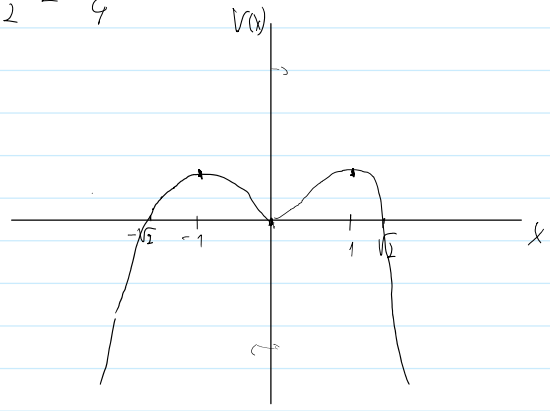
$= \frac{x^2}{2}\left(1 - \frac{x^2}{2}\right)$

(b) $E(x,y) = \frac{y^2}{2} + V(x) = C$

$V(x) = -\int_0^x -s + s^3 \, ds = \frac{s^2}{2} - \frac{s^4}{4}\Big|_0^x = \frac{x^2}{2} - \frac{x^4}{4}$

note that , $\frac{\partial V}{\partial x} = x - x^3 = \dot{y}$

using physics intution, we see a stable FP at $(0,0)$ and saddle FPs at $(-1,0), (1,0)$



(c) let look at $\dot{E}$

$\dot{E}(x,y) = y \cdot \dot{y} + x \cdot \dot{x} - x^3 \cdot \dot{x} = y(-x + x^3) + xy - x^3 y = -xy + x^3 y + xy - x^3 y = 0$

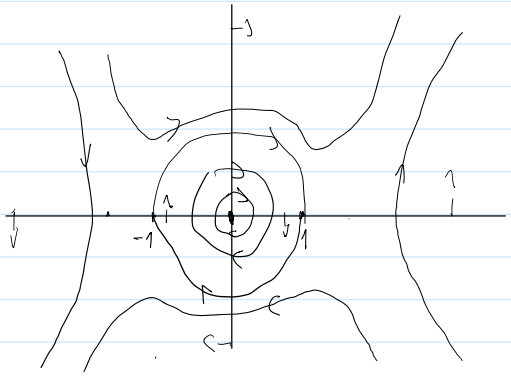
so on each trajectory $E(x,y) = C$ is preserved

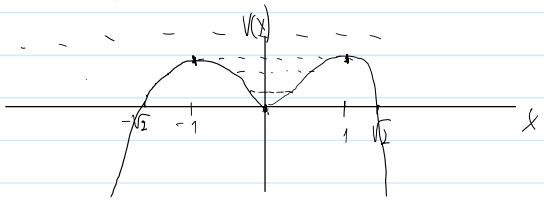
$\dot{E}(x,y) = y\cdot\dot{y} + x\cdot\dot{x} - x^2\cdot\dot{x} = y(-x+x^3) + xy - x^3y = -xy + x^3y + xy - x^3y = 0$

so on each trajetory, $E(x,y) = C$ is preserved

so $E(x,y) = \frac{y^2}{2} + \frac{x^2}{2} - \frac{x^4}{4} = C$

$y^2 = C - x^2 + \frac{x^4}{2}$





1.6 $\dot{x} = -x^3 + xy^2$

$\dot{y} = -2x^2y - y^3$

$-(-1)^3 - 2\cdot 2^2$

$-2 + 2^2\cdot 2 = -1^3$

2^4

(a) first, lets find the systems FPs

$\dot{x} = x(-x^2+y^2) = 0$ $\quad x=0$, $y^2 = x^2$

$\dot{y} = -y(2x^2+y^2) = 0$ $\quad y=0$, $2x^2+xy^2=0 \Leftrightarrow x=y=0$

so $(0,0)$ is the only FP

$L(x,y) = x^2+y^2$

for L to be a Lyapunov function, we need to show:

1. $L(x,y) > L(x^*,y^*) = L(0,0)$ for x,y around $(0,0)$
2. $\dot{L} \le 0$ ($\dot{L} < 0$ for $x,y \neq 0$ in our case).

1. $L(0,0) = 0^2+0^2 = 0$

notice that for $L(x,y) = x^2+y^2$, for any $x,y \neq 0$, $L(x,y) > 0$

2. $\dot{L} = 2x\cdot\dot{x} + 2y\cdot\dot{y} = 2x\cdot(-x^3+xy^2) + 2y\cdot(-2x^2y - y^3) = -2x^4 + 2x^2y^2 - 4x^2y^2 - 2y^4 = -2(x^4 + x^2y^2 + y^4)$

notice that $(x^4 + x^2y^2 + y^4) > 0$ for all $x,y \neq 0$

so $\dot{L} < 0$ for all $x,y \neq 0$

so $L(x,y)$ is a Lyapunov function

(b) since we found a strong Lyapunov function for $(0,0)$ FP, it means it is asymptotically stable

(b) since we found a strong Lyapunov function for (0,0) FP, it means it is asymptotically stable.

since it is the only FP and since $\forall x, y \neq 0 \quad L(x,y) > L(0,0)$
all trajectories asymptotically converge to (0,0)